Employability Improvement Platform (EIP)

Technical & Business Logic Documentation

Version 0.0.1 — Current Implementation

Covers Authentication, Role Workflows, Quiz & Cohort Engines, and System-Wide Business Rules

1. Login & Authentication Implementation

The platform supports four user roles — Super Admin, College Admin, Trainer, and Student — all authenticated through a single shared login endpoint. Authentication is implemented with JSON Web Tokens (JWT), issued on successful login and delivered both as a bearer token and as an HTTP-only cookie, so the same session mechanism works for both API clients and browser-based navigation.

1.1 Authentication Flow

Every login request submits a role, an email address, and a password. The Super Admin account is a single, fixed identity configured through environment variables — it is never stored in the database and never goes through an approval workflow. Every other role (College Admin, Trainer, Student) is a real record in the users collection, created either through the self-registration form or through Google Sign-In.

- The submitted role is normalized (trimmed, lowercased) and mapped through a small alias table, so a legacy frontend value such as "knowledge_admin" is treated identically to "college_admin" without requiring any change to the shipped HTML.
- For non-Super-Admin roles, the system looks up the account by email, then verifies the password against a bcrypt hash — plaintext passwords are never stored or compared.
- The role selected at login must match the role stored on the account; a mismatch (e.g. a Trainer attempting to log in as a Student) is rejected with "Wrong role selected."
- Login attempts are rate-limited (15 per minute per client) to reduce brute-force risk.

1.2 Role Identification & Session/Token Handling

On successful authentication, the system issues a JWT whose payload carries the user's role as an additional claim, alongside the user's database ID (or the literal identity "super_admin" for the fixed admin account). This claim is what every protected route inspects to authorize access — a single role_required(...) decorator, shared across every module, rejects any request whose token role is not in the route's allowed list, returning a 403 Forbidden.

- Access tokens expire after a configurable number of hours (12 by default), set via environment configuration.
- Tokens are accepted from either the Authorization header or an HTTP-only cookie, giving the frontend flexibility without weakening security — the cookie is marked SameSite=Lax and optionally Secure, depending on environment.
- Logging out clears the authentication cookie; the token itself is stateless and simply expires on its own schedule.

1.3 Login Validation Conditions

Before an account can be used to log in, several conditions are enforced in sequence:

- Role, email, and password must all be present in the request.
- For the Super Admin, the email and password must exactly match the configured environment values; any mismatch returns a generic "Invalid email or password" — the system deliberately does not reveal which field was wrong.
- For every other role, the account must exist, the password hash must match, and the selected role must match the stored role.
- The account's approval status is then checked (see 1.4) before a token is ever issued.

1.4 Invalid Credential & Inactive/Unapproved Account Handling

College Admin, Trainer, and Student accounts are not usable immediately after registration — they enter the system with an approvalStatus of "pending" and must be approved before they can log in. This is the platform's core account-activation gate:

- Pending accounts receive: "Your account is awaiting approval" (HTTP 403) — login is blocked outright.
- Rejected accounts receive: "Your account has been rejected" (HTTP 403).
- Any other non-"approved" status is treated conservatively and blocked with a generic inactive-account message.
- Only accounts with approvalStatus = "approved" proceed to token issuance.

Approval authority is split by role: only a Super Admin can approve or reject a College Admin or a Trainer, while a Student can be approved or rejected by either a Trainer or the Super Admin. To prevent a race condition where both a Trainer and the Super Admin act on the same pending student simultaneously, the approval update is written as an atomic, conditional database operation that only succeeds while the record is still "pending" — whichever actor's request reaches the database first wins, and the second request is rejected with an explicit "already approved/rejected by another approver" conflict response, rather than silently overwriting the first decision.

1.5 Dashboard Redirection Based on Role

A successful login response includes a redirect target resolved server-side from the authenticated role — super_admin.html, college_admin.html, trainer.html, or student.html — so the frontend never has to guess or hardcode which dashboard corresponds to which role.

1.6 Security Validations Currently Implemented

- Passwords must be at least 8 characters and include an uppercase letter, a lowercase letter, a digit, and a special character, enforced by a regular expression at both registration and password-reset time.
- Email addresses and 10-digit mobile numbers are format-validated before any account is created.
- Duplicate-account protection: registration is rejected if the email already exists, and role-specific identifiers (Roll Number for students, Employee ID for trainers, TNEA Code for college admins) must also be unique.
- Email verification via a One-Time Password (OTP) is required before both registration and password reset can complete. The OTP is time-limited (10 minutes), and once verified, a separate short-lived verification token (15 minutes) must be presented to actually complete the registration or password change — the OTP itself cannot be reused directly for that final step.
- Sensitive routes (OTP send/verify, registration, login, password reset) are all rate-limited per minute to blunt automated abuse.

2. Complete User Workflow

This section traces each role's journey through the platform from login to logout, based on the routes and permissions actually implemented.

2.1 Super Admin

The Super Admin is the platform's single unrestricted role, with full visibility across every college.

- Login: authenticates using the fixed environment-configured credentials; no approval step applies.
- Dashboard: aggregated, platform-wide view — cohort counts, quiz management summaries, and analytics charts drawn from live data, never cached or hardcoded.
- College Management: creates and manages the master list of colleges and departments that every other role's dropdowns and eligibility checks are driven from.
- Trainer Management: reviews pending Trainer registrations, approves or rejects them, and assigns an approved Trainer to a specific college/department.
- Student Management: reviews pending Student registrations platform-wide and can approve or reject them (concurrently with Trainers, subject to the first-decision-wins rule in Section 1.4).
- Assessment/Quiz Management: full authority to create, edit, publish, schedule, cancel, and archive quizzes across any college, using either Manual Entry, Bulk Excel Upload, or the permanent Question Bank as the question source.
- Reports: quiz-response exports, participation and performance analytics, and section-wise breakdowns, computed platform-wide.
- Cohort Management: configures the Placement Rules (score bands and weighting) that drive every cohort assignment on the platform, and can trigger a one-time backfill to reconcile any historical cohort data.
- Marks Management: enters interview marks and performs final validation of Create-Quiz results for any student on the platform, exactly mirroring the Trainer's own marks workflow but without college scoping.
- Logout: clears the session cookie; the JWT itself simply expires on its normal schedule.

2.2 College Admin

College Admin is a deliberately read-only role, scoped to the single college the account is registered against. Every route exposed to this role is a GET request — there is no create, edit, delete, publish, or marks-entry capability anywhere in this dashboard.

- Login: subject to the same approval gate as Trainers and Students; approval is granted only by the Super Admin.
- Dashboard: lists assessments (quizzes) that are in scope for the admin's own college, including quizzes targeted at "All Colleges" and quizzes explicitly targeted at this college by name.
- Student management: view-only — College Admin can see submitted student responses for any assessment in scope, but cannot approve, edit, or delete student records; that authority belongs to Trainer/Super Admin.
- Trainer visibility: College Admin does not manage Trainers directly; Trainer approval and assignment remain exclusively a Super Admin function.
- Reports: can view per-assessment response counts and drill into individual student responses for their own college.
- Permissions and restrictions: strictly scoped to one college and strictly read-only across the entire dashboard — this is enforced in code by reusing the exact same eligibility logic the Student-facing quiz visibility check uses, so a College Admin can never see a materially different picture of "what's happening at my college" than what actually exists.

2.3 Trainer

The Trainer role is scoped to their assigned college and carries authoring and marks-entry responsibilities that College Admin does not have.

- Login: subject to Super-Admin approval; upon approval, the Super Admin also assigns the Trainer's college and department.
- Dashboard: shows quizzes and assessments relevant to the Trainer's own college.
- Quiz Creation: authors quizzes via Manual Entry (typing questions directly) or Bulk Upload (an Excel workbook), configuring cohort targeting, college targeting, scheduling, and the Section/Difficulty distribution grid described in Section 3.
- Quiz Validation: every quiz submission — regardless of authoring method — passes through the same server-side validation engine described in Section 3 before it can be saved or published.
- Student Results: reviews submitted Create-Quiz attempts for their college, enters interview marks for each result, and performs final validation once interview marks are recorded (see Section 4).
- Assessment Monitoring: views live participation and performance analytics for their own college, using the same shared analytics computation the Super Admin dashboard uses, scoped down to one college.
- Permissions: also approves or rejects pending Student registrations (alongside the Super Admin); cannot approve College Admin or Trainer accounts, and cannot act on students outside their own scope of visibility for results/marks entry.

2.4 Student

The Student journey is the platform's core assessment-taking loop.

- Login: requires prior approval by a Trainer or the Super Admin.
- Dashboard: lists assessments bucketed into Upcoming, Pending (currently attemptable), Attempted (in progress), and Completed, computed live against the current time and the student's own submission history.
- Entry Level state: every newly approved student starts with no cohort assigned — labeled "Entry Level" throughout the system — until the two-stage cohort-generation process described in Section 4 completes.
- Quiz availability: a quiz or assessment is only visible to a student if it matches that student's current cohort target, and, where the item is college/department-scoped, if the student's own college and department also match.
- Taking assessments: starting an assessment draws a fresh, independently randomized set of questions from the configured pool (see Section 3.10) and starts a deadline clock for that attempt.
- Auto submission: if a student's browser session lapses or is closed, the server itself detects that the attempt's deadline has passed the next time anything touches that attempt (loading the dashboard, resuming, or submitting) and force-submits it with whatever answers were saved — the student is never left in an indefinitely open attempt.
- Result generation: on submission, section-wise and overall scores are computed immediately and stored as a permanent, immutable snapshot — later profile changes (name, college transfer) never rewrite a result already on record.

- Cohort assignment: baseline assessment scores feed into the two-stage Entry Level → Cohort A/B/C engine (Section 4); Create-Quiz results feed into a separate, validation-driven cohort engine described in Section 4.4.
- History and reports: students can review their own quiz history, section-wise performance, and time taken per attempt.

3. Quiz Validation Logic

Every quiz — whether authored by a Trainer or a Super Admin, and regardless of whether it was built through Manual Entry, Bulk Excel Upload, or the Question Bank — passes through one shared validation function before it can be saved as a draft or published. This guarantees the two dashboards can never disagree about what makes a quiz valid.

3.1 Required Field Validation

- Quiz title is mandatory and must be unique across the platform (case-insensitive comparison); re-saving a quiz under its own unchanged title is explicitly exempted from triggering a false duplicate.
- Assessment category, start date/time, end date/time, and duration (in minutes) are all mandatory.
- The end date/time must fall strictly after the start date/time.
- Duration must be a positive whole number of minutes.

3.2 Question-Level Validation (Manual Entry and Bulk Upload)

- Question text is mandatory.
- At least two non-empty answer options are required per question.
- At least one correct answer must be selected, and it must reference a non-empty option; both index-based and letter-based (A/B/C/D) correct-answer formats are accepted and normalized identically.
- A "Single Choice" question type may not have more than one correct answer marked.
- Marks must be a positive number.
- Section and Difficulty (Easy/Medium/Hard) become mandatory the moment the quiz uses the Section Distribution grid described below; otherwise, an unrecognized difficulty value is still rejected if one is supplied.

3.3 Duplicate Question Validation

Two questions are treated as duplicates when they share the same section and the same question text, ignoring case and extra whitespace. This check runs uniformly across Manual Entry and Bulk Upload, and separately, the permanent Question Bank enforces its own duplicate fingerprint — a hash of section, normalized question text, and normalized option list — so the same question can never be stored twice in the reusable bank even if the difficulty label differs between the two submissions.

3.4 Minimum Questions & Question Count Validation

At least one question must be present in the pool for Manual Entry/Bulk Upload quizzes (Question Bank mode draws its pool live from the bank instead). "Questions Available" is never trusted from the client — it is always recalculated server-side from the actual stored question count, so it can never drift from what was really authored or uploaded.

3.5 Section Distribution & Difficulty Distribution Validation

The current Create-Quiz wizard uses a granular grid — Available and Display counts, broken out per section and per difficulty (Easy/Medium/Hard). Three rules apply to every cell in that grid:

- Display can never exceed Available for any (section, difficulty) combination.
- The Available count configured for a cell must exactly match the number of questions actually entered or uploaded for that exact section-and-difficulty combination — not more, not less.
- Every authored or uploaded question must fall into a configured cell; a question tagged with a section/difficulty that has no matching configuration is flagged as an error rather than silently ignored.

A simpler, flat legacy format (a single Questions Displayed number, plus optional section and difficulty distributions that must each sum to that number) is also supported for backward compatibility, and is cross-checked against the actual question pool the same way.

3.6 Excel (Bulk Upload) Validation Process

A single master Excel workbook can populate the entire question pool in one upload. Each worksheet in the file represents one section (for example, Communication, Programming, Reasoning), and the header row is matched against a set of recognized aliases (e.g. "Question", "Option A"/"OptionA"/"A", "Correct Answer") so minor header naming differences do not cause the upload to fail. Every row parsed from the sheet is then run through the same question-level and duplicate validations described above before anything is written to the database.

3.7 Question Bank Mode Validation

Question Bank mode is available to the Super Admin only; a request to use it from any other role is rejected outright. In this mode the Available/Display grid is validated against the live count of questions currently sitting in the permanent Question Bank rather than against a freshly typed pool, and the actual question set for the quiz is drawn — freshly and randomly — from the bank only at publish time, never while the quiz is still a draft, so a draft can never lock in a random draw prematurely.

3.8 Applicable Cohort Validation

A quiz's cohort target must be one of a fixed set of values: a specific cohort (A, B, or C), Entry Level, or All. Any other value is rejected. This target is later used, unchanged, by the student-facing eligibility check (Section 3.10).

3.9 College Selection Validation

When a quiz targets specific colleges, the authoritative college names are always resolved server-side from the submitted college IDs — never trusted directly from the client. This matters because student eligibility is matched against college names, not IDs; storing raw IDs where names were expected would silently make a college-scoped quiz invisible to every otherwise-eligible student. Quizzes left at "All Colleges" skip this check entirely and are visible platform-wide (subject to cohort targeting).

3.10 Quiz Scheduling & Status Determination

A quiz's status is never stored — it is computed fresh from the current server time on every read, so it can never go stale:

- Draft — the quiz has been saved but not yet published.
- Scheduled — published, but the current time is still before the configured start time.
- Active (Live) — published, and the current time falls between the start and end time.
- Completed — published, and the current time is after the end time.
- Cancelled — manually cancelled by an authorized user; this overrides the normal time-based cycle.
- Archived — manually archived after completion or cancellation; this overrides every other state.

Editing rules follow directly from this status: a quiz is fully editable only while it is Draft or Scheduled, and becomes permanently locked the instant its start time passes. Deletion is allowed only while still a Draft; a Scheduled quiz can instead be Cancelled. Archiving is only available once a quiz is Completed or Cancelled.

4. Cohort Calculation Logic

The platform implements two distinct, independently-triggered cohort engines that both write to the same authoritative field on a student's account. Understanding why there are two, and how they interact, is essential to understanding cohort behavior anywhere else in the system.

4.1 Entry Level

Every newly registered student begins with no cohort value at all, displayed and treated throughout the system as "Entry Level." A student remains in Entry Level for as long as neither of the two conditions below (baseline assessment plus interview, or a validated Create-Quiz result) has completed. While in Entry Level, a student is only shown quizzes and assessments whose cohort target is "Entry Level" or "All" — nothing scoped to Cohort A, B, or C is visible to them yet.

A student leaves Entry Level the moment either cohort-generation path below successfully assigns a real cohort (A, B, or C) to their account; once that happens, the assignment does not automatically re-fire from the same engine again — a student cannot be silently reassigned to a different cohort just by having the same engine's trigger condition re-checked.

4.2 Assessment Process — Baseline Assessment + Manual Interview

This is the platform's original, dedicated cohort-generation path, driven from the general Assessment engine (separate from the Create-Quiz feature described in Section 3):

- Risk/Baseline Assessment: when a student completes an assessment flagged as the "baseline" type, their scored percentage is stored on their account as the Baseline Assessment Score. This alone does not yet assign a cohort.
- Manual Assessment/Interview: a Trainer or Super Admin separately records an Interview Score (0–100) for the student. This is a manually entered score, not derived from any quiz.
- Cohort generation only proceeds once BOTH the Baseline Assessment Score and the Interview Score exist on the student's record. Until then, the student stays in Entry Level indefinitely, however long that takes.

4.3 Final Score Calculation — Weighted Combination

Once both required inputs are present, a Final Employability Score is computed as a weighted average:

Final Employability Score = (Assessment % × Assessment Weight + Interview % × Interview Weight) ÷ (Assessment Weight + Interview Weight)

Both weights are read fresh from a Placement Rules configuration document every time this calculation runs — nothing is hardcoded in the application logic. If the Super Admin has never configured Placement Rules, the system seeds a sensible one-time default (60% Assessment weight, 40% Interview weight) exactly once; from that point forward, every calculation reads whatever is currently configured, so a Super Admin's edit to the weighting takes effect immediately across the whole platform with no code changes and no need to recompute historical students.

Example: a student with an Assessment percentage of 80 and an Interview score of 70, under the default 60/40 weighting, produces a Final Employability Score of (80 × 0.6 + 70 × 0.4) ÷ 1.0 = 76.0.

4.4 Cohort Assignment — Threshold/Range Logic

The Final Employability Score is mapped to a cohort letter using score bands that are, again, read fresh from the same Placement Rules configuration, not hardcoded. The default seed values are:

Cohort	Score Range (Default)	Meaning
Cohort A	75 – 100	Highest employability readiness
Cohort B	50 – 74.999	Moderate employability readiness
Cohort C	0 – 49.999	Needs foundational improvement

If a score falls outside every configured band (for example, a gap left by a misconfigured range), the system does not fail silently — it falls back to the lowest-bound cohort rather than leaving the student unassigned.

4.4.1 A Second, Independent Cohort Path — Create-Quiz Validation

Separately from the baseline-assessment-plus-interview engine above, the Create-Quiz feature (Section 3) has its own cohort-assignment path, because quiz results there go through a Trainer/Super Admin marks-and-validation workflow rather than an automatic assessment score:

- After a student submits a Create-Quiz attempt, its percentage score is stored, and the result enters a "Interview Pending" status.
- A Trainer or Super Admin enters interview marks (0–100) for that specific result. The moment marks are entered, a Final Average is immediately computed as (Quiz Percentage + Interview Marks) ÷ 2, and a cohort is derived from that Final Average using the same Placement Rules bands described above. The result status becomes "Interview Completed."
- A Trainer or Super Admin then formally Validates the result (only possible once interview marks exist), moving its status to "Validated."
- Validating a result is the point at which this engine commits a cohort to the student's actual account: the student's cohort is recalculated as the average Final Average across every one of their Validated Create-Quiz results — not just the single most recent one — and that average is written to the student's account as their authoritative cohort, exactly as the baseline-assessment engine writes its own result.
- Precedence rule: once a student's cohort has been set by the baseline-assessment-plus-interview engine, the Create-Quiz validation path will not overwrite it. This prevents an unrelated Create-Quiz result from silently reassigning a cohort that was deliberately set by the platform's original, single-purpose cohort mechanism. The Create-Quiz engine only ever sets or updates a cohort that either doesn't exist yet or was itself last set by this same Create-Quiz path.

4.5 What Happens After Multiple Assessments — Recalculation Over Time

For the Create-Quiz path, cohort assignment is not a one-time event: every time an additional Create-Quiz result is validated for the same student, their cohort is fully recomputed as the average Final Average across all of their Validated results to date, and the account-level cohort field is updated to match. A student who moves from a lower cohort into a higher one — or vice versa — as their body of validated results grows will see their account-level cohort (and therefore which future quizzes they are eligible for) shift accordingly. The baseline-assessment engine, by contrast, is a single, one-time calculation per student — it does not currently recompute if the student were to be assessed again.

Separately from the users collection, every cohort write from either engine is also mirrored into a dedicated student_cohort record, carrying the student's roll number, current cohort, the score that produced it, and which engine set it — purely as an additive, query-friendly reporting shadow of the same authoritative data, never a second source of truth in its own right.

5. Business Logic & Internal Calculations

5.1 Assessment / Quiz Score Calculation

For both the general Assessment engine and the Create-Quiz feature, a submitted attempt's score is computed as Marks Obtained ÷ Total Marks × 100, rounded to two decimal places, both overall and broken down per section. Every question in the current question bank is worth one mark, so obtained marks and correct-answer count are numerically identical in that context, though the calculation is expressed generically in terms of marks so it continues to work correctly if per-question marks are ever varied.

5.2 Final Employability Score & Cohort Recalculation

Covered fully in Section 4.3–4.5: a weighted average of assessment and interview percentages (baseline engine), or a running average of Final Averages across all Validated results (Create-Quiz engine), both driven by live Placement Rules configuration rather than hardcoded constants.

5.3 Average Score Calculation (Dashboards & Reports)

Wherever an "average score" is displayed — Quiz Analytics, per-student result summaries — it is computed live as the arithmetic mean of the relevant set of percentage scores at query time, never cached or pre-aggregated, so it always reflects the latest submissions.

5.4 Quiz Statistics & Dashboard Metrics

The shared Quiz Analytics computation (used identically by both the Super Admin and Trainer dashboards) derives, live from stored attempts:

- Students Attempted vs. Students Yet to Attempt, scoped to the same college/cohort/department filters as the rest of the report, measured against the full eligible student population rather than just those who happened to search or filter.
- Average, highest, and lowest score across all matching submitted attempts.
- A Top-10 Leaderboard, ranked purely by percentage score among submitted attempts matching the current filters.
- Section-wise performance, computed by re-deriving each quiz's own section structure and each attempt's section-level scoring.

5.5 Student Progress & Completion Percentage

A student's dashboard buckets each assessment into Upcoming, Pending, Attempted, or Completed based purely on time windows and the student's own submission/in-progress records — there is no separate, pre-stored "progress percentage" field; completion state is always derived live from actual attempt records and the current time.

5.6 Quiz History Statistics

A student's own quiz history view computes their average, highest, and lowest percentage across their own submitted attempts, using the exact same rounding and aggregation approach as the platform-wide analytics, just scoped to one student.

5.7 Time-Taken & Auto-Submission Calculations

Time taken per attempt is measured as the elapsed time between when the attempt was started and when it was submitted (whether submitted manually or auto-submitted by the server). Every in-progress attempt carries a deadline equal to the earlier

of a fixed default duration (30 minutes) or the assessment's own configured availability window close — whichever cuts the attempt off sooner is authoritative.

6. Conditions & Validation Summary

A consolidated view of the business rules enforced across the platform, gathered from the modules described above.

6.1 Login & Access Conditions

- Role, email, and password are all required to attempt a login.
- Only approvalStatus = "approved" accounts may log in; pending and rejected accounts are explicitly blocked with distinct messages.
- The role selected at login must match the account's stored role.
- Super Admin is a fixed, single, environment-configured identity — never a database record, never subject to approval.

6.2 User Access Permissions by Role

- Super Admin: unrestricted, platform-wide read/write access to every module, including Placement Rules configuration.
- College Admin: strictly read-only, strictly scoped to their own assigned college; no create/edit/delete/marks-entry capability anywhere.
- Trainer: authoring and marks-entry rights scoped to their own assigned college; can approve/reject Student accounts (not College Admin or Trainer accounts).
- Student: can only view and attempt content that matches their current cohort and, where scoped, their own college and department.

6.3 Quiz Eligibility Rules

- A quiz's cohort target of "All" makes it visible to every student regardless of cohort.
- A quiz's cohort target of "Entry Level" is visible only to students with no cohort assigned yet.
- A quiz's cohort target of A/B/C is visible only to students whose account cohort field currently matches exactly.
- An empty college/department scope on a quiz or assessment means it applies to every college/department; a non-empty scope requires an exact match against the student's own college and department.

6.4 Cohort Eligibility & Assessment Completion Requirements

- Cohort generation via the baseline-assessment engine requires both a Baseline Assessment Score and an Interview Score to be present before it fires.
- Cohort assignment via the Create-Quiz engine requires interview marks to be entered before a result can be Validated, and only a Validated result contributes to the account-level cohort recalculation.
- Once a cohort is set by one engine, that engine will not silently re-fire and overwrite it, and the Create-Quiz engine specifically will not override a cohort already set by the baseline-assessment engine.

6.5 Score Generation Rules

- Percentage scores are always Marks Obtained ÷ Total Marks × 100, rounded to two decimal places.
- The Final Employability Score and the Create-Quiz Final Average are both weighted/averaged combinations of assessment and interview inputs, using weights and thresholds sourced live from Placement Rules — never hardcoded in application code.

6.6 Dashboard Visibility & Student Restrictions

- A student cannot re-attempt an assessment they have already submitted.
- A student cannot continue an in-progress attempt past its deadline; the server force-submits it automatically the next time it is touched.
- An assessment whose availability window has closed, but which the student never attempted, is shown as Completed rather than silently disappearing.

6.7 Trainer & Super Admin Permissions Recap

- Only a Trainer or Super Admin may enter interview marks or validate a Create-Quiz result.
- Only a Super Admin may approve/reject College Admin and Trainer accounts, configure Placement Rules, or use Question Bank–mode quiz creation.
- A quiz becomes permanently locked from editing/deletion the instant its scheduled start time passes, for every role, without exception.

7. Overall System Flow

The following end-to-end flow shows how the modules described in this document connect together, from account creation through to reporting.

Step 1 — User Registration / Account Creation

A prospective Student, Trainer, or College Admin submits the registration form, completing email OTP verification, unique-identifier checks, and password strength validation. The account is created with approvalStatus = "pending" and, for students, a cohort of null (Entry Level).

Step 2 — Login

Once approved by the appropriate authority (Super Admin for College Admin/Trainer, Trainer or Super Admin for Student), the user logs in with role, email, and password, receiving a JWT scoped to their role.

Step 3 — Role Identification & Dashboard Access

The token's role claim determines which dashboard the user is redirected to and which API routes they are authorized to call; a shared role_required decorator enforces this on every protected route across every module.

Step 4 — Quiz Creation

A Trainer or Super Admin authors a quiz via Manual Entry, Bulk Excel Upload, or (Super Admin only) the Question Bank, configuring cohort target, college scope, scheduling, and the Section/Difficulty distribution grid.

Step 5 — Quiz Validation

The shared validation engine checks every mandatory field, cross-checks the Section/Difficulty grid against the actual question pool, screens for duplicates, and rejects the submission with specific, per-question error messages if anything fails.

Step 6 — Quiz Publishing

A validated quiz is published, after which its status becomes time-driven (Scheduled → Active → Completed) and it becomes permanently locked from further edits once its start time passes.

Step 7 — Student Participation

Eligible students (matching cohort, college, and department scope) see the quiz on their dashboard, start it to receive a freshly randomized question set, and either submit manually or are auto-submitted by the server once their deadline passes.

Step 8 — Assessment Evaluation

On submission, section-wise and overall percentage scores are calculated immediately and stored as an immutable snapshot of the result.

Step 9 — Score Calculation

For general assessments, a scored "baseline" attempt stores a Baseline Assessment Score on the student's account. For Create-Quiz results, a Trainer/Super Admin subsequently enters interview marks, producing a Final Average.

Step 10 — Cohort Assignment

Once the required inputs exist (baseline score + interview score, or a Validated Create-Quiz result), the appropriate cohort engine computes a Final Employability Score/Final Average against live Placement Rules and writes the resulting cohort (A/B/C) to the student's account, moving them out of Entry Level.

Step 11 — Reports Generation

Super Admin and Trainer dashboards draw on the same shared analytics computation to produce participation counts, average/highest/lowest scores, leaderboards, and section-wise breakdowns — always computed live from current data, filtered by college, cohort, department, and quiz as needed, and exportable for further use.

Together, these steps form a closed loop: account approval gates access, quiz validation gates content quality, cohort logic gates which content a student can even see, and the reporting layer gives every administrative role a live, mutually consistent view of the same underlying data — with no module maintaining its own disconnected copy of the rules that govern any of the others.